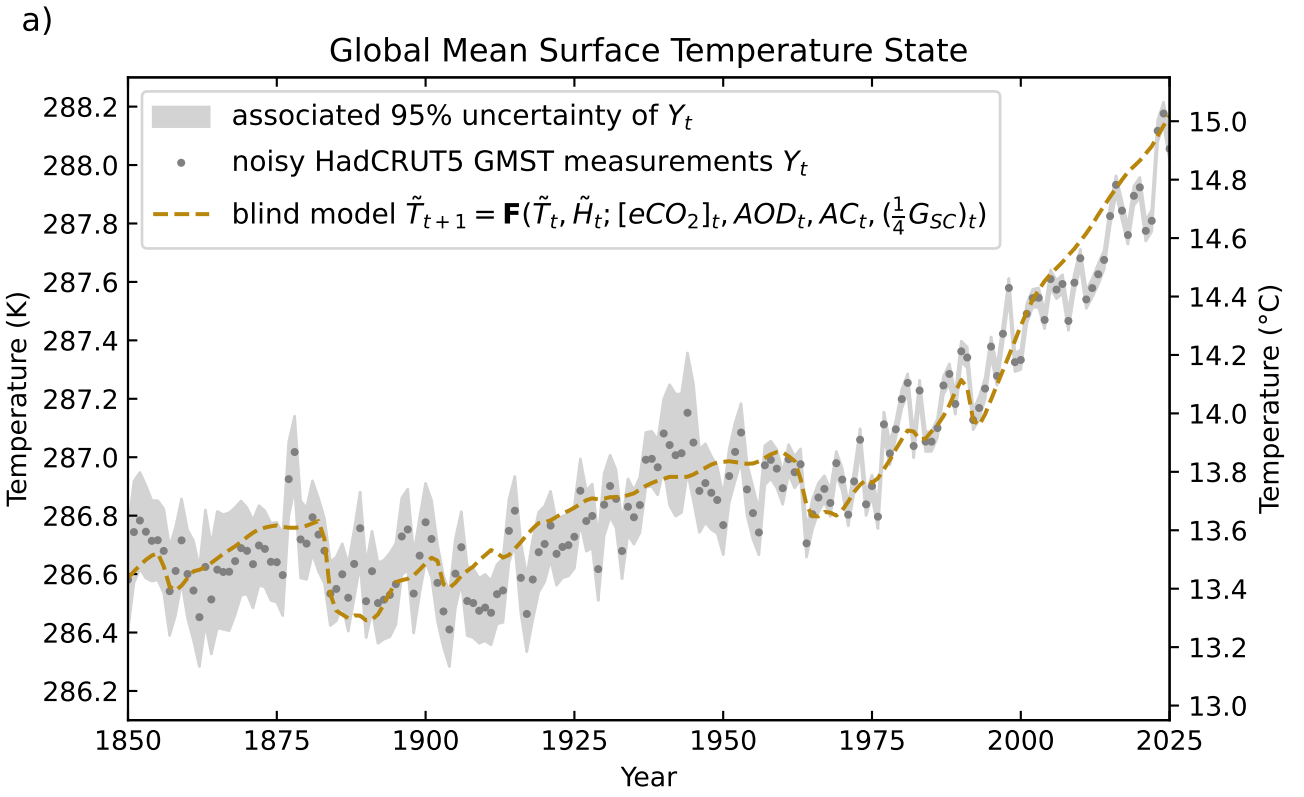
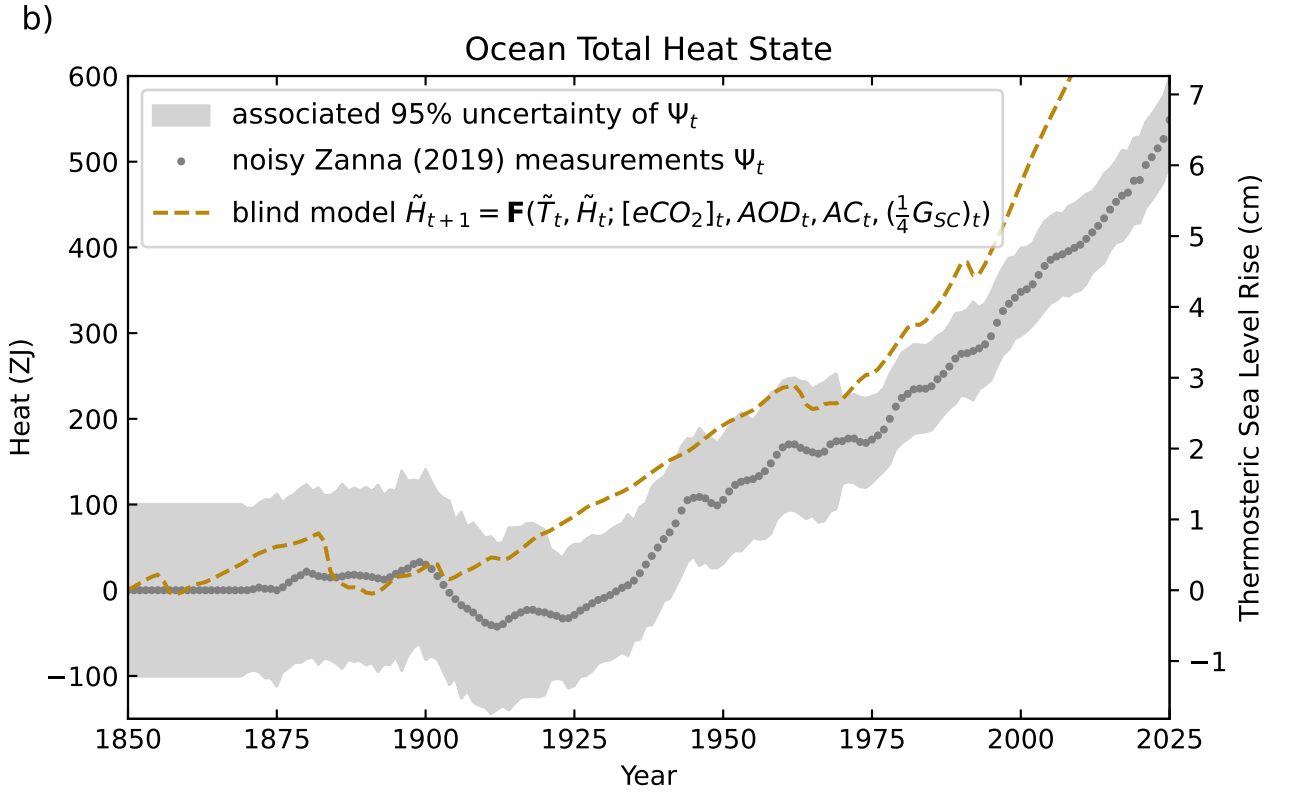

a)
Global Mean Surface Temperature State
associated 95% uncertainty of Y_t
noisy HadCRUT5 GMST measurements Y_t
blind model $\tilde{T}_{t+1} = \mathbf{F}(\tilde{T}_t, \tilde{H}_t; [eCO_2]_t, AOD_t, AC_t, (\frac{1}{4}G_{SC})_t)$
Temperature (K)
Temperature (°C)
Year
288.2
288.0
287.8
287.6
287.4
287.2
287.0
286.8
286.6
286.4
286.2
15.0
14.8
14.6
14.4
14.2
14.0
13.8
13.6
13.4
13.2
13.0
1850
1875
1900
1925
1950
1975
2000
2025


b)
Ocean Total Heat State
associated 95% uncertainty of Ψ_t
noisy Zanna (2019) measurements Ψ_t
blind model $\tilde{H}_{t+1} = \mathbf{F}(\tilde{T}_t, \tilde{H}_t; [eCO_2]_t, AOD_t, AC_t, (\frac{1}{4}G_{SC})_t)$
Heat (ZJ)
Thermosteric Sea Level Rise (cm)
Year
600
500
400
300
200
100
0
−100
7
6
5
4
3
2
1
0
−1
1850
1875
1900
1925
1950
1975
2000
2025